Light

GCSE Computer Science — Boolean Logic (Set 2)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

00:00 Start Pause Reset

← Back to Boolean Logic

Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A vending machine dispenses a drink when a coin is inserted OR a contactless card is tapped.
What type of logic operation does this describe?

- **A.** AND
- **B.** OR
- **C.** NOT
- **D.** EQUALS

Q2 (1 mark)

A safe opens only when the correct key is used AND the correct code is entered.
What type of logic operation does this describe?

- ○ **A.** NOT
- ○ **B.** OR
- ○ **C.** AND
- ○ **D.** XOR

Q3 (1 mark)

A light automatically turns ON when it is NOT daytime.
If daytime = 1 (true), what is the output for the light?

- ○ **A.** 0 (light is OFF)
- ○ **B.** 1 (light is ON)
- ○ **C.** 2 (light is dim)
- ○ **D.** Daytime

Q4 (1 mark)

An AND gate has inputs R = 1 and S = 1.
What is the output?

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** 2
- ○ **D.** RS

Q5 (1 mark)

An OR gate has inputs J = 1 and K = 1.
What is the output?

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** 2
- ○ **D.** JK

Q6 (1 mark)

A NOT gate has input T = 1.
What is the output?

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** T
- ○ **D.** NOT

Q7 (1 mark)

A logic circuit has 3 inputs: E, F, and G.
How many rows are needed in the truth table to show all possible combinations?

- ○ **A.** 3 rows
- ○ **B.** 6 rows

- ○ **C.** 8 rows
- ○ **D.** 16 rows

Q8 (1 mark)

A lift moves when the up button OR the down button is pressed.
The up button is pressed but the down button is not pressed.
Will the lift move?

- ○ **A.** No, because only one button is pressed
- ○ **B.** No, because both buttons must be pressed
- ○ **C.** Yes, because at least one button is pressed
- ○ **D.** Yes, because no buttons are pressed

Q9 (1 mark)

A game character can attack when stamina is full AND a weapon is equipped.
Stamina is full but no weapon is equipped.
Can the character attack?

- ○ **A.** Yes, because stamina is full
- ○ **B.** Yes, because one condition is met
- ○ **C.** No, because both conditions must be met
- ○ **D.** No, because weapons are optional

Q10 (1 mark)

In Boolean algebra, which symbol represents the OR operation?

- A. ∧
- B. ∨
- C. ¬
- D. ×

Q11 (1 mark)

In Boolean algebra, which symbol represents the AND operation?

- A. ∨
- B. ∧
- C. +
- D. −

Q12 (1 mark)

Which of the following correctly represents NOT R in Boolean notation?

- A. R + 1
- B. R − 1
- C. $\bar{R}$ or ¬R
- D. R × 0

Q13 (2 marks)

Evaluate the Boolean expression: (NOT R) AND S
Where R = 0 and S = 1.

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** R
- ○ **D.** S

Q14 (2 marks)

Evaluate the Boolean expression: J OR (K AND L)
Where J = 0, K = 1, L = 0.

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** K
- ○ **D.** JKL

Q15 (2 marks)

A truth table is created for a circuit with 2 inputs.
The first row shows inputs E=0, F=0.
What inputs should be shown in the second row?

- ○ **A.** E=0, F=0
- ○ **B.** E=0, F=1

- **C.** E=1, F=1
- **D.** E=1, F=0

Q16 (2 marks)

Here is part of a truth table:

R	S	Output
0	0	0
0	1	?
1	0	0
1	1	1

The circuit is an AND gate. What value replaces the question mark?

- **A.** 0
- **B.** 1
- **C.** R
- **D.** S

Q17 (2 marks)

Here is part of a truth table:

J	K	Output
0	0	0
0	1	1

1	0	?
1	1	1

The circuit is an OR gate. What value replaces the question mark?

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** J
- ○ **D.** K

Q18 (2 marks)

An ATM dispenses cash when:
(Card is valid AND PIN is correct) OR Emergency override is active
A user inserts an invalid card with the correct PIN. Override is not active.
Will the ATM dispense cash?

- ○ **A.** Yes, because the PIN is correct
- ○ **B.** Yes, because one AND condition is met
- ○ **C.** No, because card AND PIN must both be valid, and override is not active
- ○ **D.** No, because the override blocks all transactions

Q19 (2 marks)

A student claims: 'Applying NOT twice to any value returns the original value.'
Is this statement correct?

- ○ **A.** No, NOT NOT 0 = 0 but NOT NOT 1 = 0
- ○ **B.** No, NOT can only be applied once
- ○ **C.** Yes, NOT NOT 0 = 0 and NOT NOT 1 = 1
- ○ **D.** Yes, but only for the value 1

Q20 (2 marks)

A 2-input AND gate has a complete truth table with 4 rows.
How many rows have an output of 0?

- ○ **A.** 1 row
- ○ **B.** 2 rows
- ○ **C.** 3 rows
- ○ **D.** 4 rows

Q21 (2 marks)

A 2-input OR gate has a complete truth table with 4 rows.
How many rows have an output of 1?

- ○ **A.** 1 row
- ○ **B.** 2 rows
- ○ **C.** 3 rows
- ○ **D.** 4 rows

Q22 (2 marks)

Evaluate: E AND (NOT F)
Where E = 0 and F = 0.

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** E
- ○ **D.** F

Q23 (2 marks)

A smart home system turns on the heating when:
Temperature is LOW AND (NOT Summer)
The temperature is LOW and it is Summer.
Will the heating turn on?

- ○ **A.** Yes, because temperature is LOW
- ○ **B.** Yes, because at least one condition is true
- ○ **C.** No, because NOT Summer = 0 when it is Summer
- ○ **D.** No, because heating cannot run in any season

Q24 (3 marks)

Complete the truth table for: E OR F OR G

E	F	G	Output
0	0	0	?
0	0	1	1

0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

What value replaces the question mark?

- **A.** 0
- **B.** 1
- **C.** 3
- **D.** EFG

Q25 (3 marks)

A logic circuit implements: (NOT R) OR S
For which input combination is the output 0?

- **A.** R=0, S=0
- **B.** R=0, S=1
- **C.** R=1, S=0
- **D.** R=1, S=1

Q26 (3 marks)

A robot moves forward when:

• Path is clear AND battery is charged, OR
• Manual override is enabled
Which Boolean expression correctly represents this?

- ○ **A.** Path AND Battery AND Override
- ○ **B.** (Path AND Battery) OR Override
- ○ **C.** Path OR Battery OR Override
- ○ **D.** (Path OR Battery) AND Override

Q27 (3 marks)

A logic circuit has the following truth table:

E	F	Output
0	0	1
0	1	0
1	0	0
1	1	0

Which Boolean expression matches this truth table?

- ○ **A.** E AND F
- ○ **B.** E OR F
- ○ **C.** NOT (E OR F)
- ○ **D.** NOT (E AND F)

Q28 (3 marks)

Evaluate the Boolean expression: (R OR S) AND (S OR T) AND (R OR T)
Where R = 1, S = 0, T = 0.

- ○ **A.** 0
- ○ **B.** 1
- ○ **C.** RST
- ○ **D.** 3

Q29 (3 marks)

A traffic light system uses sensors: North (N), South (S), and Emergency (E).
The North-South road gets a green light when: (N OR S) AND (NOT E)
Which scenario gives a green light to the North-South road?

- ○ **A.** N=0, S=0, E=0
- ○ **B.** N=1, S=0, E=1
- ○ **C.** N=0, S=1, E=0
- ○ **D.** N=1, S=1, E=1

Q30 (3 marks)

A logic circuit implements: R AND S AND T
In a complete 3-input truth table, how many rows have an output of 0?

- ○ **A.** 1 row
- ○ **B.** 4 rows
- ○ **C.** 7 rows

○ **D.** 8 rows

Total: **55** marks • Time: **30 min**

↑ Back to top

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